BLK II - UNIT 5: METOC Mission Products
a. Mission Preparation

The goal of the PC will be to show you have the capability to use critical thinking skills to identify the thresholds and considerations for different mission types. To successfully demonstrate competency in this objective, you will be given a series of questions and tasks designed to display your ability to correctly identify the thresholds and weather considerations for the given missions. This element to the METOC career field will be demonstrated with the use of the approved reference material. For the remainder of this objective, follow your instructor's guidance as they provide instruction for completing the requirements for this objective.

Mission Types, Thresholds, and Considerations

Mission Types

Depending on the assets and specific mission at your unit, you may have to support various types of missions. If you are stationed at an airfield, you may spend most of your time supporting flying missions that care about surface weather for takeoff and landing, as well as upper-level hazards like icing and turbulence. However, if you are supporting an Army unit, you may focus more on ground-based missions and surface weather impacting the maneuverability of troops and equipment. No matter your assignment, it is important that you become familiar with all the types of missions that you will be supporting.

Mission Thresholds

Every mission type has limiting factors that can prevent the mission from being executed. As a METOC forecaster, you must use the mission specific thresholds along with your forecast to determine the impact your forecast has on a specific mission. The most common way you can communicate potential mission impacts is by using a color-coded system called a "stoplight chart" that can quickly alert mission planners to potential weather hazards impacting their planning process.

A stoplight chart contains the different weather features that can impact certain missions, and information on the different levels of those features and to what degree they impact the mission.

Weather Feature	Favorable	Marginal	Unfavorable
Sky Condition	Clear / Scattered	Broken / Overcast >020	Broken / Overcast <020
Thunderstorms	> 20 sm	10 - 20 sm	< 10 sm
Visibility	> 6 sm	3 - 6 sm	< 3 sm

Flight Missions

Flight missions involve various roles such as Intelligence, Surveillance & Reconnaissance (ISR), transport, and search & rescue (SAR). These missions use aircraft ranging from large military transports to smaller drones for tasks such as mapping, environmental monitoring, and supporting ground troops, often leveraging their ability to cover large areas efficiently.

Key Mission Areas:

- **ISR:** Collecting intelligence, identifying threats, and tracking enemy movements.
- **Transport:** Moving personnel, cargo, and equipment.
- **Strike & Interdiction:** Providing air-to-ground and air-to-air combat support.
- **Search & Rescue (SAR):** Locating and delivering supplies to people in distress, even if they can't pull them from water (Coast Guard).

Fixed Wing

A fixed-wing aircraft uses stationary wings that generate aerodynamic lift as the craft moves forward. Unlike helicopters with rotating blades, these aircraft rely on engines for propulsion, offering greater range and efficiency for long-distance travel, and encompass various types of aircraft such as small civilian airplanes, large military planes, and fixed-wing drones.

Fixed Wing Aircraft Mission Thresholds			
Parameter	**Favorable**	**Marginal**	**Unfavorable**
Sky Condition	Clear / Scattered	Broken / Overcast > 030	Broken / Overcast < 030
Icing	None	Trace / Light	Moderate / Severe
Turbulence	None	Light	Moderate to Extreme
Thunderstorms	> 25 sm	5 - 25 sm	< 5 sm
Precipitation	None / Light	Moderate	Heavy
Surface Visibility	> 5 sm	2 - 5sm	< 2sm

Rotary Wing

A rotary wing aircraft uses spinning blades for lift, allowing it to take off and land vertically, hover in place, and fly in any direction. The most common example of a rotary wing aircraft is a helicopter like the UH-60 Blackhawk. These versatile aircraft are crucial for military, emergency, and civilian tasks where traditional runways aren't feasible.

Rotary Wing Aircraft Mission Thresholds			
Parameter	**Favorable**	**Marginal**	**Unfavorable**
Sky Condition	Clear / Scattered	Broken / Overcast >020	Broken / Overcast <020
Icing	None	Trace / Light	Moderate / Severe
Turbulence	None	Light	Moderate to Extreme
Thunderstorms	> 20 sm	10 - 20 sm	< 10 sm
Precipitation	None	Light / Moderate	Heavy
Surface Visibility	> 6 sm	3 - 6 sm	< 3 sm

Ground Missions

Ground missions are typically focused on the trafficability, or the movement and/or maneuvering of troops or equipment from one location to another. These missions are focused on surface and sub-surface-based impacts rather than upper-level impacts such as icing or turbulence. Troop safety is crucial for mission success, therefore the weather parameters that affect trafficability the most are:

- **Temperature:** Affects troops by requiring adjusted hydration, clothing, and movement timelines.

- **Wind:** Impacts visibility, and equipment function.

- **Visibility:** Low visibility can impact movement, targeting, and limit air support targeting ability.

- **Precipitation / Flash Flooding:**
    - Flooding caused by intense precipitation, often temporarily overwhelming drainage infrastructure.

- **Soil Type:**
    - The smaller the grain of the soil, the stronger it will be when dry.

    - When wet, the smaller the grain, the more hazardous to traffic the soil will become.

    - The best example here is clay. Packed clay has a microscopic grain size, and when dry, can tolerate thousands of vehicles a day or more without any deterioration. When wet and agitated, the clay will quickly form slippery, heavy mud

- **Soil Moisture:** Relative Soil Moisture (RSM) is a ratio representing the current amount of water in the soil compared to its total water-holding capacity, typically represented as a percentage with 0% indicating completely dry, to 100% being completely saturated. You can utilize the RSM Model charts in Bifrost to determine RSM levels.

        - Map > Add Layer > Model Data > Environment > LIS Near Real Time > Relative Soil Moisture LIS NRT.

    - Exceptionally dry soil (RSM 0-5%) poses less risk to trafficability but creates other tactical concerns to visibility via dust formation.

    - Damp soil (RSM 20-40%) is the best for trafficability with moisture being high enough to prevent dust but not saturated enough to break down into mud under vehicle stress.

    - Moist to saturated soils pose a high risk to trafficability with low soil strength, due to the high moisture content.

- **Types of Vehicles:**
    - The type and size of the vehicles will also impact on the duration that the land is being traversed.

    - Tracked vehicles (tanks) will have more impact on the upper soil layers but generally have lower ground pressure than wheeled vehicles (Trucks). Wheeled vehicles will exert more pressure on the ground and may generate ruts.

- The weight of a vehicle will also impact the terrain, so where several light vehicles may pass, a single main battle tank could destroy a potential path.

- **Frequency of Use:**
  - This relates to the soil strength as many vehicles passing in a short time will introduce more stress to the soil than the same number passing over a longer time.
  - The spacing of the convoy matters, for example on most modular bridging systems, only one vehicle may be mounted on the bridge at once.

Ground Mission Thresholds			
Weather Feature	**Favorable**	**Marginal**	**Unfavorable**
Surface Winds	< 15KT	15 - 25KT	> 25KT
Max Temperature	50 - 75F	75 - 90F	> 90F
Min Temperature	50 - 75F	50 - 32F	< 32F
Thunderstorms	> 25 sm	10 - 25 sm	< 10 sm
Relative Soil Moisture	< 40%	40 - 70%	> 70%
Surface Visibility	> 3 sm	2 - 3 sm	< 2 sm

Coastal / Oceanic Missions

Coastal and Oceanic missions include any mission having to do with coastal or open water areas. These missions are like ground mission in that they care more about surface impacts rather than upper air impacts. In addition to the typical surface-based features impacting ground missions, coastal and oceanic missions must consider the impacts of water features on the safety and functionality of the vehicle and crew on the water.

Coastal Missions

Coastal missions involve a broad range of operations including maritime law enforcement, port security, defense readiness, and aids to navigation. These roles blend military, law enforcement, regulatory, and humanitarian functions to protect America's coasts, ports, and maritime domain.

Key Mission Areas:

- **Maritime Law Enforcement:** Impeding drug smuggling, human trafficking, and enforcing federal laws.
- **Ports, Waterways, & Coastal Security (PWCS):** Protecting the Maritime Transportation System from terrorism, sabotage, and espionage, and responding to attacks.
- **Defense Readiness:** Conducting military operations, port security, and providing support for national defense.
- **Aids to Navigation (ATON):** Maintaining lighthouses, buoys, and Vessel Traffic Services for safe commerce.

Oceanic Missions

Oceanic missions involve broad maritime operations like sea control, power projection, deterrence, and security, ranging from large-scale fleet exercises to specific efforts like combating illegal fishing and transnational crime.

Key Mission Areas:

- **Sea Control & Power Projection:** Dominating sea lanes, launching air/ground attacks from the sea, and maintaining presence.
- **Deterrence:** Preventing conflict through visible strength, including nuclear deterrence.
- **Maritime Security:** Combating piracy, drug trafficking, and enforcing fisheries laws.

Coastal / Oceanic Mission Thresholds			
Weather Feature	**Favorable**	**Marginal**	**Unfavorable**
Surface Winds	< 20KT	20 - 30KT	> 30KT
Thunderstorms	> 25 sm	5-25 sm	< 5 sm
Surface Visibility	> 4 sm	1 - 4 sm	< 1 sm
Wave Height	< 5 ft	5 - 8 ft	> 8 ft
Sea Surface Temperature	> 70F	70 F - 50 F	< 50 F
Sea State	< 4	4 - 6	> 6

Rules of Engagement (ROE) and Course of Action (COA)

Rules of Engagement (ROE)

Rules of Engagement are directives issued by competent military authority that specify the circumstances and limitations under which forces will initiate or continue missions. ROE are legal, policy-driven constraints on military action.

Key characteristics:

- ROE defines when, where, and how force can be used. It may impose restrictions on certain weapons, tactics, or targets.

- ROE ensures that military actions align with domestic and international law, as well as national policy goals.

- ROE can change depending on the operational environment. They can be very restrictive in peace operations but less so during high-intensity conflict.

- While ROE helps achieve political and military objectives by controlling escalation, they do not dictate how to achieve the mission. They only set the boundaries.

Course of Action (COA)

A mission Course of Action is the broad, potential solution a military commander develops to achieve the mission objective. It is a comprehensive plan detailing the resources, tactics, and movements of forces to accomplish the mission.

The METOC role in the planning process is crucial for mission planners to adjust their COA if unfavorable conditions exist for the current planned time or location of the mission. Effective and timely communication is of upmost importance when forecasting for missions. The information you deliver can completely alter the current mission plan or even cancel the mission entirely. It is important to note that your role in the planning process **DOES NOT** include giving suggestions or opinions on COAs. Your role is to provide the requested information and any follow-up information the mission planner might need to adjust the COA of the mission accordingly.

Key characteristics:

- A COA is an action plan designed to solve a military problem.
- Mission planners may develop multiple COAs during the planning process, evaluating and comparing them before one is approved.
- The COA is developed based on the commander's intent, the mission statement, and battlefield analysis.
- The planning process includes developing a COA that can adapt to changing circumstances as well as account for unplanned contingencies.

ROE and COA Interaction

The relationship between COA and ROE is a constant consideration during military planning. COA development is limited by ROE. When developing a COA, planners must ensure that all proposed actions are permissible under the ROE. The existence of strict ROE might force a commander to develop a less aggressive or more politically sensitive COA.

For example, peace operations with very restrictive ROE will result in a different COA than a high-intensity combat scenario with more permissive ROE. Tactical feedback can change ROE. Input from tactical-level commanders executing a COA can inform mission planners if the current ROE is unworkable or too restrictive. This can lead to an update of the ROE, which in turn might alter the approved COA.

Practice: Mission Type, Threshold, and Considerations

PC: Mission Type, Thresholds, and Considerations